Summary in Graph

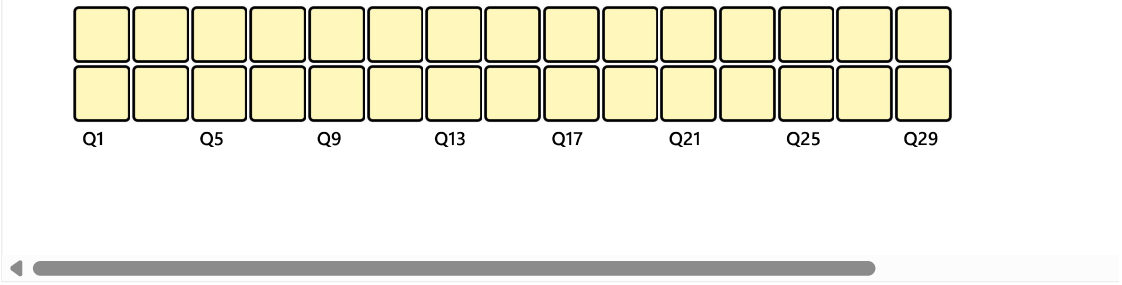
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Subject Wise Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	90 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (18) out of 175 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Which of the following properties, if present in an activation function, CANNOT be used in a neural network?

A. The function is periodic

B. The function is monotonic

C. The function is unbounded

D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Which of the following is true about the vanishing gradient problem? (Mark all that apply)

A. Tanh is usually preferred over sigmoid because it doesn't suffer from vanishing gradients

B. Vanishing gradient causes deeper layers(closer to output) to learn more slowly than earlier layers(closer to Input).

C. Leaky ReLU is less likely to suffer from vanishing gradients than sigmoid.

D. Xavier initialization can help prevent the vanishing gradient problem

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Which of the following activation functions can lead to vanishing gradients?

A. ReLU

B. Tanh

C. Sigmoid

D. None of the above

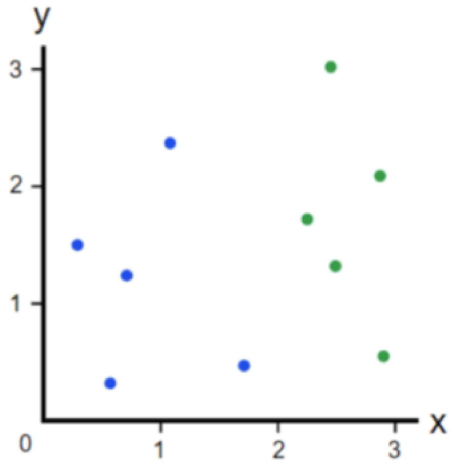
Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

If we randomly pick a datapoint, it's either blue (50%) or green (50%).

Now, we randomly classify our datapoint according to the class distribution. Since we have 5 of each color, we classify it as blue 50% of the time and as green 50% of the time.

What's the probability we classify our datapoint incorrectly?



The Dataset

Your Answer: | Correct Answer: 0.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Given d-dimensional data $\{\mathbf{x}_i\}_{i=1}^{N}$, you run principle component analysis and pick P principle components. Can you always reconstruct any data point $\mathbf{x}_i$ for $i \in \{1 \dots N\}$ from the P principle components with zero reconstruction error ?

A. Yes, if $P < d$

B. Yes, if $P = d$

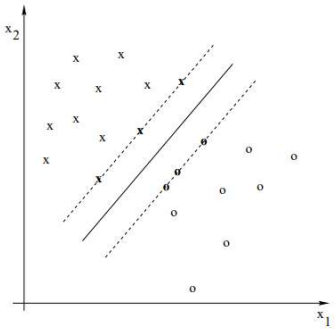
C. Yes, if $P < n$

D. No, always

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

What is the leave-one-out cross-validation error estimate for maximum margin separation in the following figure? (we are asking for a number)



Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

In a soft-margin support vector machine, if we increase C, which of the following are likely to happen?

A. The margin will grow wider.

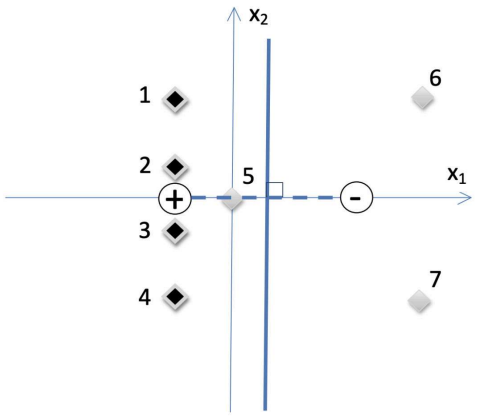
B. Most nonzero slack variables will shrink.

C. There will be more points inside the margin.

D. The norm $|w|$ will grow larger.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning



The removal of which sample will change the decision boundary in Hard-Margin SVM?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the two-class problem where class label $y \in \{T, F\}$ and each training example X has 2 binary attributes $X_1, X_2 \in \{T, F\}$. How many parameters will you need to know/estimate if you are to classify an example using the Naïve Bayes classifier ?

A. 5

B. 8

C. 3

D. 7

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

For a linear regression model, start with random values for each coefficient. The sum of the squared errors is calculated for each pair of input and output values. A learning rate is used as a scale factor, and the coefficients are updated in the direction towards minimizing the error. The process is repeated until a minimum sum squared error is achieved or no further improvement is possible. What is this process called?

A. LASSO

B. Gradient Descent

C. Least squares

D. Regularization

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

We want to predict whether a person prefers vanilla or chocolate ice cream based on a single feature: their age. We suspect that the ages of vanilla-lovers and chocolate-lovers are normally distributed with the same covariance matrix, so we build a classifier with linear discriminant analysis (LDA) and a $0-1$ loss function. Our survey of 13 random people turns up 8 vanilla lovers and 5 chocolate lovers of the following ages- Vanilla: $[21, 26, 27, 28, 30, 30, 31, 31]$ Chocolate: $[15, 18, 21, 22, 24]$ $\text{Determine the decision boundary}$

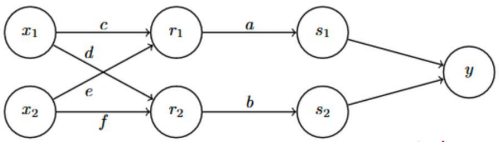
Your Answer: | Correct Answer: 23.785 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Below is a neural network with weights a, b, c, d, e, f. The inputs are x_1 and x_2. The first hidden layer computes $r_1 = max(c \cdot x_1 + e \cdot x_2, 0)$ and $r_2 = max(d \cdot x_1 + f \cdot x_2, 0)$.. The second hidden layer computes $s_1 = \frac{1}{1+exp(-a \cdot r_2)}$ and $s_2 = \frac{1}{1+exp(-b \cdot r_2)}$. The output layer computes $y = s_1 + s_2$. Note that the weights a, b, c, d, e, f are indicated along the edges of the neural network here.

Suppose the network has inputs $x_1 = 1, x_2 = -1$. The weight values are
$a = 1, b = 1, c = 4, d = 1, e = 2, f = 2$.

Calculate the values of derivative $\frac{\partial y}{\partial d}, \frac{\partial y}{\partial a}, \frac{\partial y}{\partial e}, \frac{\partial y}{\partial f}$



A. $\frac{\partial y}{\partial d} = 0.20$

B. $\frac{\partial y}{\partial a} = 0.21$

C. $\frac{\partial y}{\partial e} = -0.105$

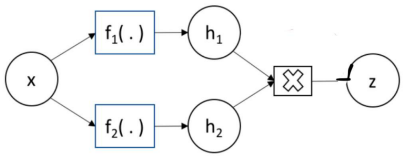
D. $\frac{\partial y}{\partial f} = 0$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Above is a deep neural nework with input x. Values x, h_1, h_2, z are all scalers.

$h_1 = f_1(x), h_2 = f_2(x), z = h_1 h_2$



Which of the following options correctly repersent the expression of $\frac{\partial z}{\partial h_1}$

A. h_1

B. h_2

C. $h_1 h_2$

D. None

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose you are training neural networks on 100 × 100 images to predict 5 classes. Neural network A consists of a single linear layer followed by a softmax output activation. Neural network B consists of a sequence of layers with dimensions 128, 512, and 32, respectively, followed by a softmax output activation. Assuming that both neural networks are trained using an identical procedure (e.g. batch size, learning rate, epochs, etc), and neither contains hidden activations, what can you generally expect about the relative performance of A and B on the test data?

A. A will outperform B

B. B will outperform A

C. A and B will perform roughly the same.

D. Not enough information to determine

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose we have a deep feedforward neural network with two layers. The first layer has four artificial neurons, and the second layer has one neuron. We want to use the network to perform binary classification on the data that is not linearly separable. This means that the dataset cannot be separated with a reasonable

performance by just using a linear classifier. Which of the following settings can help us achieve our task well?

A. Use the tanh activation function for all neurons and use the cross-entropy loss.

B. Use the identity activation function for all neurons and use the cross-entropy loss.

C. Use the tanh activation function for all neurons and use the squared error loss Sigmaid.

D. Use the identity activation function for all neurons and use the squared error loss.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Let's say you have a MLP network which is trained on 5-dimensional input data. The network has 3 hidden layers, with $50, 30$, and 20 nodes respectively. The network is trained to predict a 10-way classification task using softmax activation. If we ignore bias terms, how many weights does this neural network have?

A. $2,550$

B. $1,500,000$

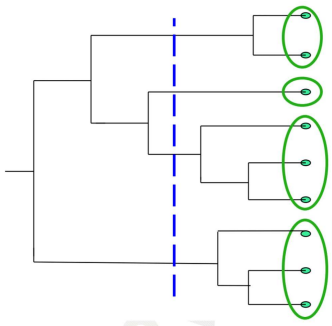
C. 115

D. $30,000$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Find the number of clusters for the given cut



Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider six points in 2D, some of them overlap. We start with three clusters (rather than from scratch):

Cluster $1 = (0,0), (1,1), (2,2)$

Cluster $2 = (0,1), (1,0)$

Cluster $3 = (0, 0)$

With Euclidean distance and complete linkage, which clusters are merged in the next step ofhierarchical clustering?

A. Cluster 1 and Cluster 2

B. Cluster 1 and Cluster 3

C. Cluster 2 and Cluster 3

D. Not enough information to determine

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Perform k-means clustering on six points $x_i = i$ for $i = 1 \ldots 6$ with two clusters. Initially the cluster centers are at $c_1 = 1, c_2 = 6$. Run k -means until convergence. What is the reduction in distortion ?

A. 2

B. 4

C. 6

D. 8

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider performing K-Means Clustering on a one-dimensional dataset containing four data points: $\{5, 7, 10, 12\}$ using $k = 2$, Euclidean distance, and the initial cluster centers are $c_1 = 3.0$ and $c_2 = 13.0$.

(I) What are the initial cluster assignments? (That is, which examples are in cluster c_1 and which examples are in cluster c_2 ?)
(II) What is the value of distortion for the clusters computed in (I) ?

(III) What are the new cluster centers after making the assignments in (I) ?

Select the correct option for statements (I), (II) and (III) respectively.

A. $C_1 = \{5, 7\}$ and $C_2 = \{10, 12\}, \quad 30, \quad C_1 = \{6\}$ and $C_2 = \{11\}$

B. $C_1 = \{10, 12\}$ and $C_2 = \{5, 7\}, \quad 29, \quad C_1 = \{11\}$ and $C_2 = \{6\}$

C. $C_1 = \{5, 7\}$ and $C_2 = \{10, 12\}, \quad 30, \quad C_1 = \{12\}$ and $C_2 = \{22\}$

D. $C_1 = \{10, 12\}$ and $C_2 = \{5, 7\}, \quad 29, \quad C_1 = \{22\}$ and $C_2 = \{12\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

A decision tree is trained on the following 2D dataset, where $(x_{i,1}, x_{i,2}, t_i)$ represents the features and the class label:

$$(-2, 2, 0), (2, 1, 0), (7, -1, 0), (9, -7, 0), (0, -6, 1), (5, -2, 1), (8, -5, 1)$$

(I) Calculate the Gini index at the root node (without split).

(II) Calculate the weighted average Gini impurity after splitting the root node on $x_2 < -1.5$.

Select the correct option for statement (I) and (II) respectively.

A. $0.489,\ 0.214$

B. $0.214,\ 0.489$

C. $0.511,\ 0.786$

D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the following design matrix, representing four sample points $X_i \in \mathbb{R}^2$.

$$X = \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}$$

We want to represent the data in only one dimension, so we turn to principal components analysis (PCA).

Compute the unit-length principal component directions of X, and state which one the PCA algorithm would choose if you request just one principal component. Please provide an exact answer, without approximation.

A. $\left[1/\sqrt{2} \quad , 1/\sqrt{2}\right]^\top$

B. $\left[1/\sqrt{2} \quad , 1/-\sqrt{2}\right]^\top$

C. $\left[1/-\sqrt{2} \quad , 1/\sqrt{2}\right]^\top$

D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a training set in $\mathbb{R}^2$ shown in the table below. Suppose we use the Perceptron algorithm for classification. We record the total number of times each point occurs in the update step (meaning the point was misclassified).

Assume the initial weight vector is $w^{(0)} = [-3, 2, 1]$, where the last term corresponds to the offset (bias) term with feature $x_3 = 1$.

x_1	x_2	y	times misclassified
-3	2	+1	0
-1	1	+1	0
-1	-1	-1	2
2	2	-1	1
1	-1	-1	0

What is the equation of the separating line expressed in terms of x_1 and x_2 determined by the algorithm?

A. $-3x_1 + 2x_2 + 1 = 0$

B. $-3x_1 + 2x_2 - 2 = 0$

C. $3x_1 - 2x_2 - 2 = 0$

D. $-x_1 + x_2 - 1 = 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

After the i-th iteration of online perceptron learning, you have a model h_i and you receive a new instance X_{i+1} . You find out that your current model misclassifies the instance as $h_i(X_{i+1}) = +1$ when you receive the actual label $Y_{i+1} = -1$. You update the model using the perceptron algorithm and get a classifier $h_{i+1} \cdot h_{i+1}$ is guaranteed to classify X_{i+1} correctly as -1? (Please enter 1 for True and 0 for False).

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Now you will run 10-fold cross-validation for training k-NN. For each of candidate k values, you train k - NN on all dataset but one of the 10 folds, then measuring an approximate validation error on the examples in that heldout fold.

When you have 5 different candidate k values to decide the best model, you will train k-NN total $\mathrm{N1}$ times. The performance of individual models (with a specific k value) will be evaluated by the mean of $\mathrm{N2}$ validation errors.

A. $\mathrm{N1} = 45, \mathrm{N2} = 10$

B. $\mathrm{N2} = 45, \mathrm{N1} = 50$

C. $\mathrm{N1} = 50, \mathrm{N2} = 10$

D. $\mathrm{N1} = 10, \mathrm{N2} = 50$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Given a K-class dataset containing N points, where sample points are described using D discrete features with each feature capable of taking V values, how many parameters need to be estimated for Naïve Bayes Classifier ?

A. $V^D K$

B. K^{V^D}

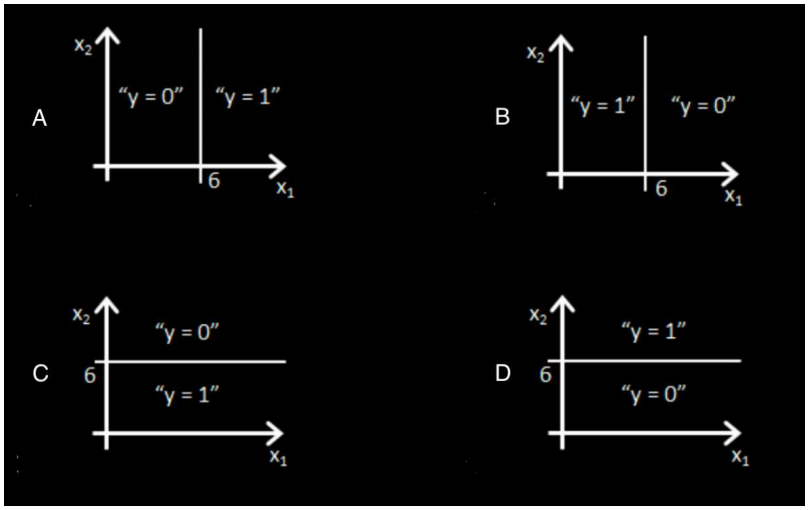
C. $VDK + K$

D. $K(V + D)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose you train a linear classifier $h_\theta(x) = sign(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$. Suppose
$\theta_0 = 6, \theta_1 = -1,$
$\theta_2 = 0$
. Which of the following figures represents the decision boundary found by your classifier?



Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the least squares regression model, $\hat{\mathbb{Y}} = \mathbb{X}\theta$. Assume that $\mathbb{X}$ and $\mathbb{Y}$ refer to the design matrix and true response vector for our training data.
Let $\hat{\gamma}$ be the parameter vector that minimizes mean squared error without regularization. Specifically:

$$\hat{\gamma} = \arg\min_{\gamma} \frac{1}{n} \|\mathbb{Y} - \mathbb{X}\gamma\|_2^2$$

Let $\hat{\beta}$ be the parameter vector that minimizes mean squared error with L_2 regularization, using a non-negative regularization hyperparameter λ (i.e. ridge regression). Specifically:

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{n} \|\mathbb{Y} - \mathbb{X}\beta\|_2^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

As we increase λ, the model variance of the model $\hat{\mathbb{Y}} = \mathbb{X}\hat{\beta}$ tends to

A. increase

B. stay the same

C. decrease

D. None of these

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Given a design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and labels $\mathbf{y} \in \mathbb{R}^n$, which of the following techniques could potentially decrease the empirical risk on the training data (assuming the loss is the squared error) ?

A. Adding the feature " 1 " to each data point.

B. Adding polynomial features to each data point.

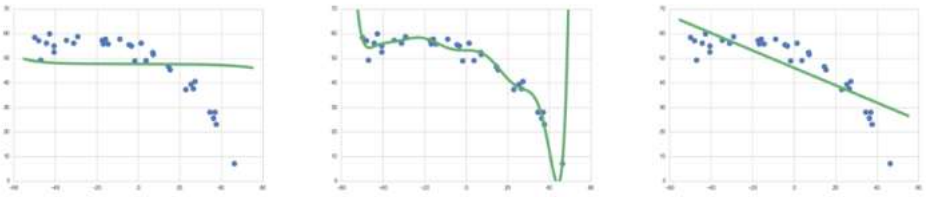
C. Centering the vector $\mathbf{y}$ by subtracting the mean $\bar{y}$ from each component y_i.

D. Penalizing the model weights with L_2 regularization.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We have fit several models depicted as curves in the following plots:



(a) (b) (c)

Select the plot that best matches each of the models below. Each plot is used exactly once.

1. Linear regression model
○ (a) ○(b) ○(c)

2. Linear regression with degree 10 polynomial features
○ (a) ○(b) ○(c)

3. Ridge regression with degree 10 polynomial features and substantial regularization.
○ (a) ○(b) ○(c)

Select the correct option :

A. c, b, a
B. a, b, c
C. b, c, a
D. c, a, b

Your Answer: **Correct Answer: A** Not Attempted Time taken: 00min 00sec Discuss

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